Vehicle Rental SaaS — Customer & Document Management Specification

Production specification covering customer profiles, drivers, identity verification, document management, rental history, payments, deposits, contracts, communication, blacklisting, privacy, and customer lifecycle management.

1. Customer Module — Main Sections

#	Section	Main purpose
1	Basic Customer Profile	Identity and contact information
2	Address & Contact	Residential, emergency and communication details
3	Identity / KYC	Customer verification and identity checks
4	Documents	Licence, ID, passport and other customer documents
5	Drivers	Primary customer and additional authorized drivers
6	Rental Eligibility	Age, licence and business eligibility rules
7	Rental History	Bookings, rentals, returns and incidents
8	Payments & Deposits	Customer payments, deposits, refunds and balances
9	Contracts & Signatures	Rental agreements and signed documents
10	Communication	Email, SMS, WhatsApp and in-app communication
11	Notes & Activity	Internal notes, timeline and audit history
12	Blacklist / Restrictions	Controlled restrictions and risk flags
13	Review & Verification	Approval workflow before rental/online booking

2. Basic Customer Information

- Customer type: Individual / Business / Organization
- First name
- Middle name
- Last name
- Preferred/display name
- Date of birth
- Gender where the business legitimately requires it
- Nationality
- Customer ID / internal reference number
- Profile photo where appropriate
- Customer status: Active, Pending Verification, Suspended, Blacklisted, Archived
- Customer creation date and source

3. Contact & Address Information

- Primary phone number
- Secondary phone number
- Email address
- Preferred communication channel
- Country
- Province/state

- District/city
- Street/address
- Postal/ZIP code where applicable
- Emergency contact name
- Emergency contact relationship
- Emergency contact phone
- Billing address if different
- Business/company address for corporate customers

4. Customer Account & Online Portal

- Customer login/account
- Email verification status
- Phone verification status
- Password/authentication state
- Last login
- Account creation date
- Marketing communication preference
- Notification preferences
- Language preference
- Online booking enabled/disabled
- Account security/session history

5. Identity Verification / KYC

- Verification status: Not Started, Pending, Verified, Rejected, Expired
- Verification method
- Identity document type
- Identity document number
- Name matching result
- Date of birth matching result
- Document expiry validation
- Manual verification by staff
- Verified by
- Verified at
- Verification notes
- Reverification requirement
- Do not store unnecessary sensitive information

6. Customer Document Management — Dedicated Subsystem

- Central customer documents area
- Document type and category
- Document number
- Issuing authority/country
- Issue date

- Expiry date
- Uploaded file
- File metadata
- Verification status
- Verified by / verified at
- Rejection reason
- Document version
- Document history
- Archive/soft delete
- Expiry reminders
- Permission-controlled access
- Audit log for upload, verification, replacement, archive and deletion

7. Customer Documents

- Citizenship / national ID
- Passport
- Driving licence
- Driving licence front/back
- Visa/residency document where applicable
- Address proof where required
- Business registration documents for corporate customers
- Tax/business identification documents where applicable
- Additional driver documents
- Signed customer forms
- Other supporting documents

8. Document Management Features

- Search and filter by customer, type and status
- Valid / Expiring Soon / Expired / Pending / Rejected states
- Secure upload and preview
- File type and size validation
- Document versioning
- Document verification workflow
- Configurable expiry reminders such as 30/15/7 days
- Bulk upload where useful
- Secure private file storage
- Temporary authorized access for downloads/previews
- Document retention/archive policy
- Document access audit trail

9. Driver Management

- Primary customer/driver
- Additional authorized drivers

- Driver name and contact information
- Driver date of birth
- Driving licence number
- Licence category/class
- Licence issue date
- Licence expiry date
- Licence verification status
- Driver relationship to customer
- Driver restrictions
- Driver-specific documents
- Driver rental history
- Driver eligibility at booking/pickup

10. Rental Eligibility

- Minimum age requirement
- Driving licence requirement
- Minimum licence holding period
- Allowed licence categories
- International licence requirement where applicable
- Customer verification requirement
- Deposit requirement
- Customer-specific rental restrictions
- Vehicle/category restrictions
- Eligibility status
- Eligibility notes
- Backend must re-check eligibility during booking and pickup

11. Rental History

- All bookings
- Completed rentals
- Active rentals
- Cancelled bookings
- No-shows
- Overdue returns
- Vehicle pickup/return history
- Rental duration
- Vehicles rented
- Rental revenue
- Damage incidents
- Late fees
- Extra kilometre/hour charges
- Customer feedback/ratings where used
- Contract history

12. Payments, Deposits & Financial History

- Total paid
- Outstanding balance
- Payment transactions
- Payment methods
- Security deposits
- Deposit holds/releases
- Refunds
- Failed payments
- Invoice history
- Credit/adjustment records
- Payment references
- Payment status
- Financial data should be permission-restricted

13. Contracts & Signatures

- Rental contracts
- Contract templates used
- Contract version
- Vehicle linked to contract
- Booking linked to contract
- Start/end dates
- Signed status
- Customer signature
- Additional driver signatures where required
- Signed contract file
- Signature timestamp
- Contract amendments
- Contract history

14. Communication & Notifications

- Email history
- SMS history
- WhatsApp messages where integrated
- In-app notifications
- Booking confirmations
- Pickup reminders
- Return reminders
- Payment reminders
- Document expiry reminders
- Contract notifications
- Marketing messages with consent controls

- Communication preferences
- Do not send marketing messages without the required consent

15. Customer Notes & Activity Timeline

- Internal staff notes
- Customer-visible notes where appropriate
- Call/contact history
- Verification actions
- Booking actions
- Payment actions
- Document actions
- Blacklist/restriction actions
- Contract actions
- Staff user who performed the action
- Timestamped activity timeline

16. Blacklist / Restrictions / Risk Controls

- Blacklist status
- Restriction type
- Reason
- Start date
- End date if temporary
- Created by
- Approved by where required
- Supporting evidence/document
- Internal notes
- Restriction severity
- Appeal/review status
- Do not expose internal risk notes to the customer

17. Customer Detail Page

- Overview
- Profile
- Contact & Address
- Verification
- Documents
- Drivers
- Bookings
- Active Rentals
- Rental History
- Payments
- Deposits
- Refunds

- Invoices
- Contracts
- Communication
- Notes
- Restrictions
- Activity Timeline
- Audit History

18. Customer Lifecycle

- Lead/Prospect if enabled
- Registered
- Email/phone verified
- Documents submitted
- Verification pending
- Verified
- First booking
- Active renter
- Returning customer
- Restricted/Suspended
- Blacklisted where justified
- Archived

19. Related Modules / Places

- Customers — master customer records
- Verification/KYC — identity and licence verification
- Documents — centralized document management
- Drivers — additional authorized drivers
- Bookings — customer reservations
- Active Rentals — current rental contracts
- Payments — customer transactions
- Deposits — security deposits
- Refunds — returned deposits and payment refunds
- Invoices — customer billing
- Contracts — rental agreements
- Notifications — reminders and transactional communication
- Marketing — coupons, loyalty, referrals and campaigns
- Support/CRM — customer inquiries and service cases
- Reports — customer revenue, retention, frequency and lifetime value
- Audit Logs — customer-data changes and access history

20. Recommended Database Entities

- customers
- customer_profiles
- customer_contacts
- customer_addresses
- customer_emergency_contacts
- customer_documents
- customer_document_versions
- customer_document_verifications
- customer_verification_cases
- customer_drivers
- driver_documents
- customer_preferences
- customer_consents
- customer_notes
- customer_activity_logs
- customer_restrictions
- customer_blacklist_entries
- customer_payment_summary
- customer_tags
- customer_segments

21. Customer Document Record — Minimum Data

Field	Purpose
id	Unique document ID
tenant_id	Tenant isolation
customer_id	Customer owner
driver_id	Optional additional-driver owner
document_type_id	Configured document type
document_number	Official document number
issuing_authority	Issuing organization/country
file_key	S3/object-storage key
file_name / mime_type / file_size	File metadata
issue_date / expiry_date	Validity dates
status	Valid/Expiring/Expired/etc.
verification_status	Pending/Verified/Rejected
rejection_reason	Reason when rejected
verified_by / verified_at	Verification audit
version	Document version
notes	Internal notes
created_at / updated_at	Timestamps
deleted_at	Archive/soft delete if used

22. Customer API / Backend Rules

- Never trust tenant_id/company_id supplied by the frontend.
- Resolve tenant from authenticated context and authorized membership.
- Every customer, driver, document, booking, payment and note query must be tenant-scoped.
- Customer identity and eligibility checks must happen on the backend.
- Do not allow a customer to access another customer's documents or rental records.
- Document downloads/previews must be authorization-protected.
- Payment webhooks and customer payment mutations must be idempotent.
- Customer deletion should normally use controlled archival/anonymization rather than unsafe hard deletion.
- Keep legally required records according to the business's retention policy.
- All sensitive customer-data access and important mutations should be auditable.

23. Privacy & Security

- Collect only data necessary for rental operations, verification, billing and legitimate business purposes.
- Restrict identity documents, financial information and internal risk notes using granular permissions.
- Encrypt data in transit and use secure storage for private documents.
- Use short-lived authorized file access rather than public permanent document URLs.
- Support customer consent/preferences for marketing communications.
- Maintain an audit trail for sensitive document and profile access.
- Do not expose internal blacklist/risk notes through the customer portal.
- Support configurable retention and archival policies.

24. Frontend Navigation

- Customers → All Customers
- Customers → Add Customer
- Customers → Verification
- Customers → Documents
- Customers → Drivers
- Customers → Rental History
- Customers → Blacklist / Restrictions
- Customer Detail → Overview / Verification / Documents / Drivers / Bookings / Rentals / Payments / Contracts / Activity
- Documents → All / Expiring / Expired / Pending Verification
- Reports → Customer Revenue / Retention / Lifetime Value / Rental Frequency

25. Add Customer — Recommended Form Flow

- 1. Customer Type
- 2. Basic Identity
- 3. Contact Information
- 4. Address
- 5. Identity / KYC
- 6. Driving Licence
- 7. Additional Drivers
- 8. Documents

- 9. Emergency Contact
- 10. Rental Eligibility
- 11. Preferences & Consent
- 12. Review & Create

26. Required vs Conditional Fields

Do not make every customer field mandatory. A normal customer may need name, contact information and required rental/verification data. Driving licence fields become mandatory when the customer will drive a vehicle. Business registration information applies to corporate customers. Additional driver fields apply only when additional drivers are added. The backend must enforce the final rules.

27. Final Customer Checklist

- ■ Basic customer identity
- ■ Phone/email
- ■ Address
- ■ Emergency contact
- ■ Identity verification
- ■ Driving licence
- ■ Required customer documents
- ■ Additional drivers if applicable
- ■ Rental eligibility
- ■ Preferences and consent
- ■ Payment/deposit linkage
- ■ Contract linkage
- ■ Review and backend validation
- ■ Activate customer account